distribution

DH Exam Questions WS 2025/2026

Topics:

1. AI methods and their application in healthcare/clinical decision support (**12/14 points**)
    a. L1 -> Intro
    b. L2 -> Clinical and ambient data
    c. L3 -> AI methods overview
    ~~d. L12 -> AI with small data (slides not available, not included in exam)~~
    e. E1-E3 (Medical data)
    f. E4-E6 (LSTMs)
2. Digital Biomarkers (**6/8 points**)
    a. L4 -> Temporal data
    b. L5 -> Mental health
3. Digital Twin (**14 points**)
    a. L7 -> Concept, intro
    b. L8 -> Body models
    c. L9 -> Modelling
    d. L10 -> Population scaling
    e. E7-E9
4. Model-guided medicine (**8/10 points**)
    a. L11 -> BN + MGM
    b. L13 -> Wisdom oriented healthcare and model science
    c. E10-E12 (Bayesian Networks)
5. Infrastructure, standards and regulations (**6 points**)
    a. L06 -> IT infrastructure
    b. L14 -> DIGA, regulations, prescription models

Total: 50 points, Time: 60 mins

Matriculation No.:

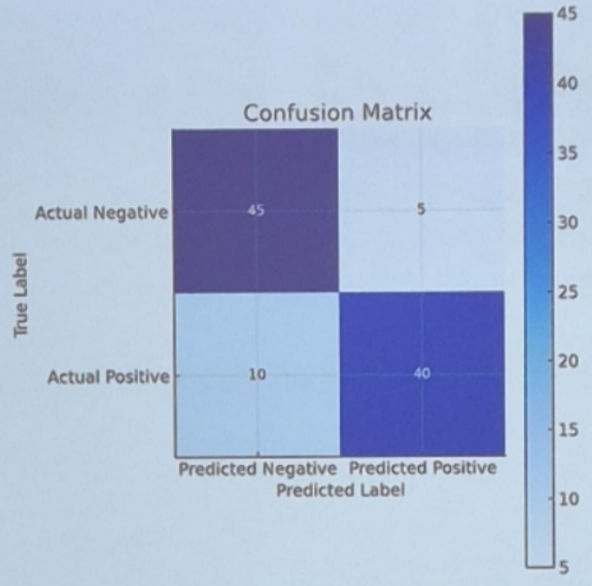
Section 1: AI methods and their applications in healthcare (12 points)

1. Complete the table by assigning one correct example of medical data from the list to the corresponding category of medical data. (3 points)

List:
[Age], [Diagnoses], [Family history], [Surgeries], [Physical activity], [Diet/nutrition], [Sleep], [Smoking], [Creatinine values], [MRI scan], [Mass spectrometry]

Category of medical data	Example
Behavioural	
Omics	
Lab results	

2. Using the confusion matrix illustrated below, answer the following questions:



(a) Calculate accuracy. (1 point)

(b) In a clinical setting, a high Recall for critical events is important because it minimizes (1 point)
(choose one: False Negatives , False Positives, F1 score, Accuracy).

3. Match the algorithm to the most appropriate data type (from list A) and the reason (from list B) combination. (3 points)

List A

A1. Discrete or categorical data.
A2. High-dimensional data.
A3. Data with small sample sizes.
A4. Longitudinal or time-series data.
A5. Probabilistic relationships among variables.

List B

B1. Employs a memory-based mechanism to capture evolving trends and handle variable-length sequences.
B2. Explicitly represents dependencies among variables using conditional probability distributions, allowing for inference under uncertainty.
B3. Learns a compressed representation that can reconstruct or generate new instances from high-dimensional input.
B4. Relies on aggregated sample efficiency to handle small datasets effectively.
B5.
Models the probability structure of discrete categories for better classification or decision-making.

AI algorithm	Data type	Reason
LSTM		
Bayesian network		
Variational Autoencoder		

4. List two types of time series data in healthcare. (1 point)

Topic 1 Example question

You are given a sequence of hourly heart rate data processed for an LSTM. The "Active" column is determined by the rule: **Active = 1 if (HR ≥90 AND Mask = 1), else 0.**

Find the missing values marked with [ ] in the table below. (2 points)

Hour	HR_hour	Mask	Δ t	Active
08:00	85	1	0	[A]
09:00	NaN	0	[B]	0
10:00	NaN	0	[C]	0
11:00	92	1	0	[D]
12:00	NaN	0	1	0

A: 0

B: 1

C: 2

D: 1

Topic 3 Example question

An Agent-Based Model is built from several core components. The table below has rows for **Agent**, **Environment**, and **Global Variables**. From **List A**, choose the definition that best fits the component. From **List B**, choose the specific example or attribute associated with it.

Note: Do not reuse items. *(3 points)*

List A (Definitions)

- A1. The spatial or network domain where interactions occur.
- A2. The aggregate outcome or pattern that arises from the simulation (e.g., the epidemic curve).
- A3. Autonomous units with individual states and behavioral rules.
- A4. System-wide settings that apply to the simulation as a whole, not specific to one individual.
- A5. The visualization interface used to watch the simulation.

List B (Examples/Attributes)

- B1. An individual's current "Health Status" (e.g., Susceptible or Infected).
- B2. A grid of patches, a city map, or a social network graph.
- B3. The total duration of the simulation (e.g., 100 days).
- B4. The "Transmission Probability" setting that affects all contacts in the model.
- B5. The calculation of R_0 after the simulation ends.

Category	Definitions	Examples/Attributes
Agent		
Environment		
Global variables		

Category	Definitions	Examples/Attributes
Agent	A3	B1
Environment	A1	B2
Global variables	A4	B4

Topic 4 Example question

Understanding dependencies in a Directed Acyclic Graph (DAG) is crucial for Bayesian Networks. The table below has rows for **Serial (Chain)**, **Diverging (Common Cause)**, and **Converging (Collider)** structures. From **List A**, choose the diagrammatic representation (A, B, C). From **List B**, choose the condition under which nodes A and C become **independent**.

Note: Do not reuse items. *(3 points)*

List A (Representation)

A1. A ← B → C

A2. A → B → C

A3. A → B ← C

A4. A ↔ B

A5. A → B → A (Cyclic)

List B (Independence Condition)

B1. A and C are independent **only if B is NOT observed** (and no descendant of B is observed).

B2. A and C become independent **if B IS observed** (B mediates the influence).

B3. A and C become independent **if B IS observed** (B explains away the correlation).

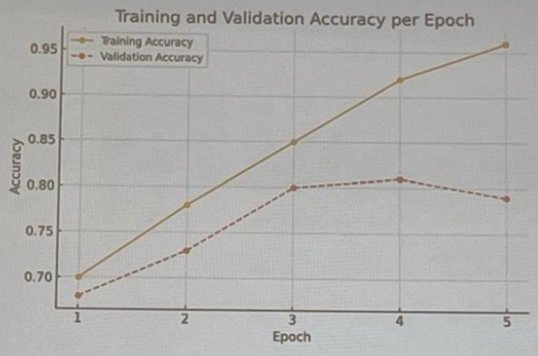
B4. A and C are always independent regardless of B.

B5. A and C are never independent.

Structure type	Representation	Independence condition
Serial (chain)		
Diverging (common cause)		
Converging (collider)		

Structure type	Representation	Independence condition
Serial (chain)	A2	B2
Diverging (common cause)	A1	B3
Converging (collider)	A3	B1

6. A healthcare team trains an LSTM model to predict patient readmissions. They observe the following training and validation accuracy after 5 epochs:



You have three key intervals to describe. Complete the table by selecting observations from List A and explanation from List B. (3 points)

List A

A1. Both training and validation accuracies are steadily improving.
A2. Training accuracy improves significantly, while validation accuracy plateaus or dips.
A3. Training and validation accuracy remain nearly identical.
A4. There is a relatively large gap between training and validation accuracy.
A5. Training and validation are both very low, indicating underfitting.

List B

B1. Indicates the model is learning generalizable patterns early on.
B2. Suggests overfitting, as the model is learning the training set more than it is generalizing.
B3. Implies data leakage, because there should normally be some difference between training and validation curves.
B4. Shows the model has not learned enough from either dataset (underfitting).
B5. Demonstrates perfect generalization with no discrepancy between training and validation results.

Intervals	Observations	Explanation
Epochs 1-3		
Epochs 3-5		
Final epoch		

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Exam question 3

(b) In a clinical setting, a high Recall for critical events is important because it minimizes (1 point)

(choose one: False Negatives , False Positives, F1 score, Accuracy).

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A potential exam question?

Intelligent Embedded Systems Lab

Scenario	Preferred Metric	Reason
Screening for rare disease	**Recall (Sensitivity)**	Avoid missing true cases (FN costly)
Confirmatory diagnosis	**Precision (PPV)**	Avoid false alarms or overtreatment
Alert/triage system	**F1 Score**	Balance between missed cases and false alarms
General model comparison	**AUC-ROC**	Threshold-independent ranking measure
Rare event prediction	**AUPRC**	More informative under class imbalance

Scenario in patient trajectory	LSTM component	Reason
Spikes in patient trajectory 1 year ago	Forget gate	Erares irrelevant past info
New abnormal lab result	Input gate	Decides what to remember from current input
Long term renal function patterns	Cell state	Long-term trend memory
Daily summary of patient stability	Hidden state	Short-term output summary
Monitoring of evolving patient state	Sequence flow	Processes temporal context

A statistical modeling approach that organizes parameters at multiple levels (e.g., patient, clinic, region) is called a . (1 point)

Study the two scenarios in the table below. For each row, choose among differential equation (DE) models and agent-based models (ABMs) according to scenario's characteristics. Also, provide one concise reason why your chosen model is the better choice. (2 points)

Note: Both the chosen model and corresponding reason needs to be correct.

Scenario	Favoured model	Reason
Only concerned with global prevelance trends and average disease dynamics.		
Small, diverse community with crucial individual/local interactions.		

6. The SEIR model divides the population into four categories. The table below has four rows - one each for S, E, I, and R. From list A, choose one item that matches the description of the compartment (S, E, I, or R) and the most appropriate description of the time invariant parameter(s) that governs the respective change from list B. (4 points)

Note: Do not reuse items from List A, each compartment uses exactly one description.

List A

A1. A population of unvaccinated, healthy individuals with no prior exposure.
A2. Individuals who tested positive for the pathogen but show no symptoms yet (latent phase).
A3. Patients with active symptoms, confirmed infection, and potential to infect others.

Exercise: Practice Terminology II: Parkinson's Disease

- Parkinson's Disease is a progressive nervous system disorder affecting movement, causing tremors, stiffness, ar slowing of movement.

	Traditional	Digital
Endpoint	Unified Parkinson's Disease Rating Scale (UPDRS): Clinician-administered assessment of motor and non-motor symptoms.	Gait and Movement Analysis Wearables: Sensors capturi on walking speed, stride len and tremor frequency.
Clinical Outcome	Symptom Severity: Changes in tremors, rigidity, and bradykinesia.	Patient-Reported Outcome (ePROs): Digital questionna assessing symptom progres
Biomarker	Dopamine Transporter Imaging: PET or SPECT scans measuring dopamine levels.	Keystroke Dynamics: Analy typing patterns to detect fin skill changes.